

10th May 2026 | CS6370

⇒ Latent Semantic Indexing

- unimode factor analysis. We look at data pts. and then check for the patterns. ~~do~~ linear discriminant analysis
- SVD →

From unimode → 2-mode factor analysis

$$A_k = U \Sigma V^T$$

$m \times n$ $m \times k$ $k \times k$ $k \times n$

term vectors
 $term \times concepts$
doc-doc
matrix

A_k is the reduced rank approximation of the original matrix A .

U & V are column orthogonal

V^T is row orthogonal

concepts are identical

some of the concepts have low significance, hence, removed ⇒ dimensionality reduction.

k -rank approximation (SVD is one of them)

- read the paper, it is inspt.

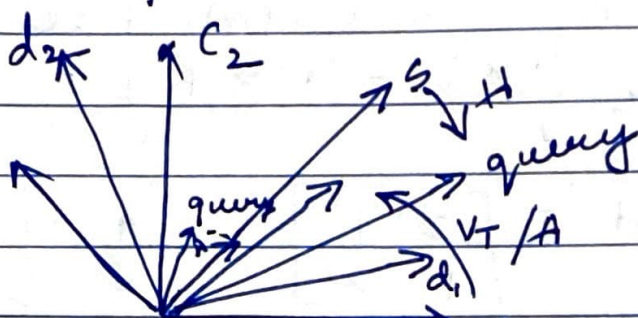
- even with $k=2$, we can see the patterns $A_k = U_2 \Sigma_2 V_2^T$, $A_k \approx A$
it has similarities.

- the dual mode adv.
 - terms & docs are both present
 - then underlying concepts can explain things more diligently.
- what things can be done -
 - compute doc. similarities
 - compute term similarities
 - compute relevance of a term w.r.t doc.
- how to determine k (in context of IR)?
 - Precision & recall
 - precision requires high k & recall req. low k .
 - also, not records the value of k .
- also, Σ is diagonal
- we should know when such $U \Sigma V^T$
 - uses on a vector, what happens to that vector
- longer matrix formⁿ (cols. orthogonal)
- aligner matrix (rows orthogonal)
- stretch matrix (stretch it, but no extraⁿ)
- $SVD = H \overset{\text{aligner stretch}}{S} A$
 - action of a matrix prod. $A \approx B \times C$ is same as the acⁿ of C

followed by the action B.

$U \rightarrow$ Hanger
 $\Sigma \rightarrow$ stretcher
 $V^T \rightarrow$ aligner

} analogy for SVD.



aligner will bring it in C the concept space
stretcher elongates it.
hanger will bring it and hang it
in the doc. space